MIL-F-2310D

17 September 1974
SUPERSEDING
MIL-F-2310C
17 September 1968

MILITARY SPECIFICATION

FRINGE, ORNAMENTAL, RAYON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an ornamental rayon fringe.

1.2 Classification. The fringe shall be of one type in the following classes, as specified (see 6.2).

Class 1 - 1-1/2 inch length.
Class 2 - 2-1/2 inch length.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-665	-	Boxes, Paperboard, Metal Edged and Components.
PPP-B-676	-	Boxes, Setup.
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods

FSC 8315

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

US Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Guide sample. Guide samples of the end item, when furnished, are solely for guidance and information to the supplier (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

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3.2 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

3.3 Material.

3.3.1 Rayon yarn. The rayon yarn used for the heading and bullion shall be 300 (± 15) denier, multifilament, bright viscose or cuprammonium rayon.

3.3.2 Cotton yarn. The cotton yarn shall be a 12/2 (± 1 count) cotton yarn.

3.4 Color. The color of the fringe shall be as specified (see 6.2). The yarns for the fringe shall be yarn dyed. The rayon yarns may also be spinning solution dyed. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished yarns shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.5. When a standard sample is not available the dyed and finished yarns shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.5.

3.4.1 Matching. The color shall match the approved standard shade under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

3.4.2 Colorfastness. The finished fringe shall show fastness to light, water, and wet drycleaning equal to or better than the standard sample when tested as specified in 4.5. When no standard sample is available, the finished fringe shall show good fastness to light, water, and wet drycleaning when tested as specified in 4.5.

3.5 Physical requirements. The finished fringe shall conform to the physical requirements of table I when tested as specified in 4.5.

TABLE I. Physical requirements

Characteristic	Class 1	Class 2
Length (overall), inches	1-1/2 (± 1/8)	2-1/2 (± 1/8)
Bullions per inch (minimum)	12	12
Heading:		
Width, inches (minimum)	1/4	3/8
Ends in float stripe (minimum)	16	24
Ends in each border (minimum)	18	28
Weight per gross yards, ounces (minimum)	72	108

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3.6 Construction.

3.6.1 Bullion. The filling yarn, which is used to form the bullion, shall be made by twisting together 2 ends of cotton yarn (see 3.3.2) and then wrapping with 3 ends of rayon yarn (see 3.3.1). The bullion shall be formed by twisting 2 picks in the same shed sufficiently so that they twist upon themselves flush up to the heading with not less than 5 turns per inch of Z-twist.

3.6.2 Heading. The heading shall be constructed from single ends of rayon (see 3.3.1) and shall be woven with a float stripe centered in the heading floating over and under 4 picks (2 bullion), with a border on each side floating over and under 2 picks (1 bullion).

* 3.7 Put up and length. Unless otherwise specified, the fringe shall be put up on commercial size flat chipboards containing 25 yards (± 2 yards). There shall be not more than three lengths per board and no one length shall be less than 5 yards. The end of the fringe shall be fastened to prevent unwinding.

3.8 Marking. Each board shall have firmly attached a white paper label printed in black with the following information:

Stock number
Nomenclature
Specification number
Supplier's name
Contract number and date
Name of contracting agency
Gross yards

3.9 Fiber identification. Each board shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act, (see 2.1).

3.10 Workmanship. The finished fringe shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels (AQL) established.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

* 4.3 First article inspection. When required (see 6.2), the first article submitted in accordance with 3.2 shall be inspected for all the provisions of this inspection applicable to end item inspection.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition to quality assurance provisions of the referenced specifications and standards, testing shall be performed on the components listed in table II for the characteristics shown. The sample unit shall be as shown in table II. The inspection level shall be level S-1. The acceptable quality level (AQL) for each characteristic shall be 6.5 test failures per 100 units.

TABLE II. Component testing

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Sample unit	Results reported as
Rayon yarn (one cone, spool or tube)	Denier, filament, luster, and type	3.3.1	1/	---	---	---
Cotton yarn (one cone, spool or tube)	Yarn count 1/	3.3.2	4021	4	One cone, spool or tube	---

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TABLE II. Component testing (cont'd)

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Sample unit	Results reported as
Filling yarn for bullion (yards)	Construction	3.6.1	Visual	1	One yard	Pass or fail

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.4.2 Inspection of the end item. Defects found during this examination shall be classified in accordance with 4.4.2.1 through 4.4.2.3.2.

4.4.2.1 Visual examination. The defects found during this examination shall be classified in accordance with the defects listed in table III. All defects shall be counted regardless of their proximity to one another, except where two or more defects represent a single local condition in the fringe, in which case only the more serious defect shall be counted. The sample unit shall be 1 yard. The AQL shall be 2.5 major and 6.5 total defects (major and minor combined) per hundred units. The sample size shall be in accordance with inspection level II. The lot size shall be expressed in units of 1 yard.

TABLE III. Classification of defects

Defect	Classification Major	Minor
Spot, stain, or soiled place 1/		X
Top edge of heading uneven (indentations exceed 1/16 inch)	X	
Top edge of heading uneven 1/ (indentations do not exceed 1/16 inch)		X
Cut, hole or tear	X	
Broken or missing warp yarn, three or more contiguous regardless of length	X	
Broken or missing bullion		X
Length of bullion uneven 1/		X
Frayed or kinked bullion		X
Crushed or wrinkled fringe		X

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TABLE III. Classification of defects (cont'd)

Defect	Classification Major	Minor
Overall length of fringe:		
Class 1		
More than 1-5/8 inches long		X
Less than 1-3/8 inches long		X
Class 2		
More than 2-5/8 inches long		X
Less than 2-3/8 inches long		X
Width of heading:		
Class 1		
Less than 1/4 inch		X
Class 2		
Less than 3/8 inch		X

1/ Clearly visible at normal inspection distance (approximately 3 feet).

4.4.2.2 Overall examination. Each defect listed below shall be counted not more than once on each board examined. The sample unit for this examination shall be one board. The sample size for this examination and the number of defects acceptable shall be as shown in table IV.

Defects

Top edge of heading uneven, continuously or intermittently over more than half the length of the piece.

Does not match specified color.

Not clean throughout.

Objectionable odor.

Identification marking missing, illegible or incorrect.

Not ticketed, labeled or invoiced in accordance with the Textile Fiber Products Identification Act.

Improperly or insecurely wound on boards, tangled, or does not unwind properly.

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TABLE IV

Lot size in yards	Sample size in boards	Acceptance number
1300 1/ or less	3	0
1,301 up to and including 3,200	5	0
3,201 up to and including 8,000	7	0
8,001 up to and including 22,000	10	0
22,001 up to and including 110,000	15	1
110,001 up to and including 550,000	25	1
550,001 and over	50	2

1/ Should the lot contain fewer than 3 boards, each board in the lot shall be examined.

4.4.2.3 Examination for length.

4.4.2.3.1 Individual boards. The boarded fringe shall be examined for length and number of individual pieces per board. Any length found to be more than the maximum specified or less than the minimum specified, and total length found to be less than the length marked on the board ticket, any board containing more than the maximum allowable number of pieces, or any one length less than 5 yards, shall be considered as a defect with respect to length. The sample size and the acceptable number of defects shall be in accordance with table IV.

4.4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of the fringe on the boards in the sample is less than the total of the lengths marked on the board tickets. The sample size for this examination shall be as shown in table IV.

* 4.4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking complies with section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Weight	Net weight of each container exceeds maximum requirements.
Content	Number of boards per intermediate package is more or less than required. 1/

1/ For this defect, two intermediate packages shall be examined from each shipping container in the sample.

* 4.5 Testing. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in table VI, shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 3 yards. The sample size shall be in accordance with table V. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

TABLE V. Sample size

Lot size in yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VI. Methods of testing

Characteristic	Requirement paragraph	Reference test method	No. determinations per indiv. sample unit	Results reported as
Labile sulfur	3.4	2020	1	Pass or fail
Colorfastness:				
Light	3.4.2	5660	1	--
Water	3.4.2	5630	1	--
Wet drycleaning	3.4.2	5622	1	--
Bullions per inch	Table I	Visual	1	To nearest whole number
Heading:				
Ends in float stripe	Table I	Visual	1	To nearest whole number
Ends in each border	Table I	Visual	1	To nearest whole number
Weight of fringe per gross (144) yards	Table I	4.6	3	To nearest 0.5 ounce per gross (144) yards
Bullion:				
Direction of twist	3.6.1	4050	1	Pass or fail
Turns per inch	3.6.1	Visual	3	Average of 3 determinations to nearest 0.5 turns
Construction of heading	3.6.2	Visual	1	Pass or fail

4.6 Weight. Three specimens in excess of one yard each shall be selected randomly from different boards in the sample. The three specimens of fringe in the relaxed state shall be cut to exactly one yard each. While in a standard

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condition as defined in FED-STD-191, each yard shall be weighed separately on a balance or scale accurate to 0.01 ounce, (a scale or balance graduated in grams may be used and results converted to ounces). The weight of the three specimens shall be averaged and this average value multiplied by 144 to determine the ounces per gross yards.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

* 5.1.1.1 Unit package. Fringe, put up on a board as specified (see 3.7), shall be completely wrapped or bagged in transparent plastic film. The wrap or bag shall be secured by pressure-sensitive transparent cellulose tape, heat sealing, or adhesive.

* 5.1.1.2 Intermediate package. Five boards of fringe, of one class and color only, unit packaged as specified, shall be placed in a snug-fitting set-up paperboard box conforming to type I, variety 1, class C, style 4 of PPP-B-676; a folding paperboard box conforming to variety 1, style III, type G, class 1 of PPP-B-566; or a metal-stayed paperboard box conforming to class 1, style A of PPP-B-665. Thumb notch requirements may be waived. The box closure shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade A of PPP-T-45, applied at the center of the length opening and extending across the bottom and up the sides of the box.

* 5.1.2 Level C (commercial packaging). Fringe, put up as specified, shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

* 5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Fringe, of one class and color only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636. The weight of the contents of each shipping container shall not exceed 65 pounds.

* 5.2.2 Level B. Fringe, of one class and color only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each container shall be closed in accordance with method II as specified in the appendix of the container specification. The weight of the contents of each shippping container shall not exceed 65 pounds.

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* 5.2.2.1 When specified (see 6.2), the shipping container shall be grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C (commercial packing). Fringe, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with US Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fringe is to be used primarily as a border on certain types of distinguishing flags.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of fringe required (see 1.2).
(c) When a first article is required (see 3.2, 4.3 and 6.4).
(d) Color required (see 3.4).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples, address the procuring office issuing the invitation for bids or request for proposal.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one properly wound board of tape. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.5 Marginal code identification. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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